

Assignment No:4

Q.1 Write a simulation program for disk scheduling using FCFS algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

55, 58, 39, 18, 90, 160, 150, 38, 184

Start Head Position: 50

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n, i, head, total = 0;
    printf("Enter total number of disk blocks: ");
    scanf("%d", &n);
    printf("Enter current head position: ");
    scanf("%d", &head);
    int req[100];
    printf("Enter number of disk requests: ");
    scanf("%d", &n);
    printf("Enter disk request string:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &req[i]);
    }
    printf("\nOrder of requests served:\n");
    printf("%d", head);
    for (i = 0; i < n; i++)
    {
        total += abs(head - req[i]);
        head = req[i];
        printf(" -> %d", head);
```

```

    }
    printf("\nTotal head movement = %d\n", total);
    return 0;
}

```

Q.2 Write a simulation program for disk scheduling using SSTF algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

98, 183, 37, 122, 14, 124, 65

Start Head Position: 53

```

#include <stdio.h>
#include <stdlib.h>
int main()
{
    int blocks, n, head;
    int req[100], visited[100] = {0};
    int i, j, min, index;
    int total = 0, distance;
    printf("Enter total number of disk blocks: ");
    scanf("%d", &blocks);
    printf("Enter number of disk requests: ");
    scanf("%d", &n);
    printf("Enter disk request string:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &req[i]);
    printf("Enter current head position: ");
    scanf("%d", &head);
    printf("\nOrder of requests served:\n");
    printf("%d", head);

```

```

for (i = 0; i < n; i++)
{
    min = 9999;
    index = -1;
    for (j = 0; j < n; j++)
    {
        if (!visited[j])
        {
            distance = abs(head - req[j]);
            if (distance < min)
            {
                min = distance;
                index = j;
            }
        }
    }
    visited[index] = 1;
    total += min;
    head = req[index];
    printf("-> %d", head);
}
printf("\n\nTotal head movement = %d\n", total);
return 0;
}

```

Q.3 Write a simulation program for disk scheduling using SCAN algorithm.

Accept total Number of disk blocks, disk request string, and current

head position from the user. Display the list of requests in the

order in which it is served. Also display the total number of head moments.

82, 170, 43, 140, 24, 16, 190, 65

Starting Head position= 50

Direction: Left

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int blocks, n, head, direction;

    int req[100], i, j, temp;

    int total = 0;

    printf("Enter total number of disk blocks: ");

    scanf("%d", &blocks);

    printf("Enter number of disk requests: ");

    scanf("%d", &n);

    printf("Enter disk request string:\n");

    for (i = 0; i < n; i++)
        scanf("%d", &req[i]);

    printf("Enter current head position: ");

    scanf("%d", &head);

    printf("Enter direction (0 = Left, 1 = Right): ");

    scanf("%d", &direction);

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (req[i] > req[j])
            {
                temp = req[i];
```

```

        req[i] = req[j];

        req[j] = temp;
    }

}

}

printf("\nOrder of requests served:\n");

printf("%d", head);

if (direction == 0)    // LEFT
{
    for (i = n - 1; i >= 0; i--)
    {
        if (req[i] < head)
        {
            total += abs(head - req[i]);

            head = req[i];

            printf(" -> %d", head);

        }
    }

    if (head != 0)
    {
        total += head;

        head = 0;

        printf(" -> %d", head);
    }

    for (i = 0; i < n; i++)
    {

```

```

        if (req[i] > head)
        {
            total += abs(head - req[i]);

            head = req[i];

            printf("-> %d", head);

        }
    }
}

else        // RIGHT
{
    for (i = 0; i < n; i++)
    {
        if (req[i] > head)
        {
            total += abs(head - req[i]);

            head = req[i];

            printf("-> %d", head);

        }
    }

    if (head != blocks - 1)
    {
        total += (blocks - 1) - head;

        head = blocks - 1;

        printf("-> %d", head);
    }

    for (i = n - 1; i >= 0; i--)

```

```

{
    if (req[i] < head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf("-> %d", head);
    }
}

printf("\n\nTotal head movement = %d\n", total);

return 0;
}

```

Q.4 Write a simulation program for disk scheduling using Look algorithm. Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of request in the order in which it is served. Also display the total number of head moments. 55, 58, 39, 18, 90, 160, 150, 38.

Start Head Position: 100

Direction: Left

```

#include <stdio.h>

#include <stdlib.h>

int main()
{
    int blocks, n, head, direction;

    int req[100], i, j, temp;

```

```
int total = 0;

printf("Enter total number of disk blocks: ");

scanf("%d", &blocks);

printf("Enter number of disk requests: ");

scanf("%d", &n);

printf("Enter disk request string:\n");

for (i = 0; i < n; i++)

    scanf("%d", &req[i]);

printf("Enter current head position: ");

scanf("%d", &head);

printf("Enter direction (0 = Left, 1 = Right): ");

scanf("%d", &direction);

for (i = 0; i < n - 1; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (req[i] > req[j])
        {
            temp = req[i];
            req[i] = req[j];
            req[j] = temp;
        }
    }
}

printf("\nOrder of requests served:\n");

printf("%d", head);
```

```

if (direction == 0)    // LEFT
{
    for (i = n - 1; i >= 0; i--)
    {
        if (req[i] < head)
        {
            total += abs(head - req[i]);

            head = req[i];

            printf("-> %d", head);

        }
    }
    for (i = 0; i < n; i++)
    {
        if (req[i] > head)
        {
            total += abs(head - req[i]);

            head = req[i];

            printf("-> %d", head);

        }
    }
}
else                    // RIGHT
{
    // Service requests to the right of head
    for (i = 0; i < n; i++)
    {

```

```

    if (req[i] > head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf("-> %d", head);

    }
}

for (i = n - 1; i >= 0; i--)
{
    if (req[i] < head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf("-> %d", head);

    }
}

printf("\n\nTotal head movement = %d\n", total);

return 0;
}

```

Q.5 Write a simulation program for disk scheduling using LOOK algorithm.

Accept total number of disk blocks, disk request string, and current head

position from the user. isplay the list of requests in the order in which it is served.

Also display the total number of head moments

15, 30, 55, 90, 125, 140, 170

Starting Head Position: 100

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int req[] = {15, 30, 55, 90, 125, 140, 170};

    int n = 7;

    int head = 100;

    int i, j, temp;

    int total = 0;

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (req[i] > req[j])
            {
                temp = req[i];
                req[i] = req[j];
                req[j] = temp;
            }
        }
    }

    printf("Order of requests served:\n");

    printf("%d", head);

    for (i = n - 1; i >= 0; i--)
    {
```

```

    if (req[i] < head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf(" -> %d", head);
    }
}

for (i = 0; i < n; i++)
{
    if (req[i] > head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf(" -> %d", head);
    }
}

printf("\nTotal head movement = %d\n", total);

return 0;
}

```

Q.6 Write a simulation program for disk scheduling using LOOK algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

86, 147, 91, 177, 45, 12, 130

Starting Head Position: 60

Direction: Right

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int req[] = {86, 147, 91, 177, 45, 12, 130};

    int n = 7;

    int head = 60;

    int i, j, temp;

    int total = 0;

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (req[i] > req[j])
            {
                temp = req[i];
                req[i] = req[j];
                req[j] = temp;
            }
        }
    }

    printf("Order of requests served:\n");

    printf("%d", head);

    for (i = 0; i < n; i++)
    {
```

```
    if (req[i] > head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf(" -> %d", head);
    }
}
for (i = n - 1; i >= 0; i--)
{
    if (req[i] < head)
    {
        total += abs(head - req[i]);

        head = req[i];

        printf(" -> %d", head);
    }
}

printf("\nTotal head movement = %d\n", total);

return 0;
}
```